Fiscal Year:	FY 2021	**Task Last Updated:**	FY 06/30/2021
PI Name:	Hada, Megumi Ph.D.		
Project Title:	Computational Model Prediction and Biological Validation using Simplified Mixed Field Exposures for the Development of a GCR Reference Field		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77446	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-RADIATION. Appendix D: Ground-Based Studies in Space Radiobiology
Start Date:	08/26/2016	**End Date:**	03/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2021 per NSSC information (Ed., 5/28/20) NOTE: End date changed to 3/31/2020 per NSSC information (Ed., 8/8/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Patel, Zarana Ph.D. (KBR/NASA Johnson Space Center) Plante, Ianik Ph.D. (KBR/NASA Johnson Space Center) Ponomarev, Artem Ph.D. (KBR/NASA Johnson Space Center) Shavers, Mark Ph.D. (KBR/NASA Johnson Space Center) Slaba, Tony Ph.D. (NASA Langley Research Center)		
Grant/Contract No.:	NNX16AR97G		
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Task Description:	This work addresses the need to develop and utilize mixed field irradiation protocols that approximately represent the shielded tissue environment in space and that can be combined with single-beam studies to validate and further improve cancer risk models. Extensive studies have been carried out on the induction of chromosomal aberrations by low- and high-LET (linear energy transfer) radiation in human lymphocytes, fibroblasts, and epithelial cells exposed in vitro. The results, which are consistent with computational modeling predictions, establish chromosome aberration models as an LET-sensitive tool for predicting damage from single ion exposures. However, there is a lack of data on chromosome aberrations induced by low dose rate chronic exposure and mixed field beams such as those expected in space. These proposed studies will define a simplified mixed field and make model predictions of the effects of dose rate, mixed fields, and shielding on expected biological damage. Chromosome aberration studies at NASA Space Radiation Laboratory (NSRL) will provide the biological validation needed to extend the computational models over a broader range of experimental conditions ((more complicated mixed fields leading up to the galactic cosmic radiation (GCR) simulator)), helping to reduce uncertainties in radiation quality effects and dose-rate dependence in cancer risk models (Cancer-3 and -4). These models can then be used to answer some of the open questions regarding requirements for a full GCR reference field, including particle type and number, energy, dose rate, and delivery order.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This strategy of single and mixed field exposures and varying dose rates along with a well-defined biological assay and established computational models will provide a unique opportunity to develop an intermediate mixed field that can closely approximate the GCR reference field. This represents a critical step towards using the GCR simulator at NSRL to reduce uncertainties in radiation quality effects and dose-rate dependence in cancer risk models.
Task Progress:	The NASA Space Radiation Laboratory (NSRL) has been upgraded to rapidly switch ions and energies to allow investigators to design irradiation protocols comprising a mixture of ions and energies representative of the galactic cosmic ray (GCR) environment. Beam selection and delivery schemes were optimized against facility and experimental constraints, and need to be validated to ensure such irradiations are a suitable representation of the space environment. Importantly, since experiments are time consuming and expensive, models capable of predicting biological outcomes over a range of irradiation conditions (single ion, sequential multi-ion, or mixed fields) are needed to confirm our understanding, and to support estimates of risk for untested complex space radiation environments. We investigated chromosome aberrations (CA) induced in human blood lymphocytes, skin fibroblast, epithelial cells, and bone marrow lymphocytes from female mice (CB6F1 mice of 100-120 days of age) that were exposed to single and mixed beams (2-ions, 3-ions, 4-ions, or 6-ions mix) of Proton, Helium, Oxygen, Silicon, Titanium, and Iron-ions at NSRL. Chromosomes from human cells were collected with the premature chromosome condensation method in the first mitosis. Chromosomes from mouse bone marrow were collected 16 months after exposure. CA yield was measured with 3-color fluorescent in situ hybridization (FISH) chromosome painting. A combination of models and programs were used to simulate CA in fibroblast cells. The code Geant4 was used to calculate the distribution of particles behind shielding after irradiation. Using calculated particle spectra, the code RITRACKS (Relativistic Ion Track) generated time-dependent differential 3D voxel dose maps, which were used in RITCARD (Radiation Induced Tracks, Chromosome Aberrations, Repair and Damage) to simulate CA in human fibroblasts. The multi-scale model described here is a major milestone toward developing computational tools to predict biological outcomes in a complex ion field environment. The simulation and experimental results are in good agreement, suggesting that the model may be highly useful in further design of a ground-based GCR simulator and for informing risk projection models in the future. Mixed beam data of human fibroblast CA were also analyzed using Incremental Effect Additivity (IEA) synergy theory, rather than the Simple Effect Additivity (SEA) synergy theory that has long been discredited for dealing with highly curvilinear Dose-Effect Relationship (DER). Results from the synergy modeling suggest neither synergy nor antagonism (NSNA).
Bibliography Type:	Description: (Last Updated: 06/26/2025)
Abstracts for Journals and Proceedings	Hada, M, Beitman AJ, Rhone JR, Zhao L, Wang S, Chang PY, Blakely, Sachs RK, Saganti PB. "Chromosome aberrations induced by simulated galactic cosmic ray mixed beam exposure in human lymphocytes, fibroblasts and epithelial cells." Presented at the 16th International Congress of Radiation Research (ICRR), Manchester, UK, August 25-29, 2019. Abstract book of ICRR 2019, August 2019. , Aug-2019
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